Charlotte Soeurng

Minneapolis, MN | charlottesoeurng@outlook.com | GitHub | Portfolio

Education

University of St. Thomas, St. Paul, MN **Graduated May 2026**

- Bachelor of Science in **Computer Science** | Minor in **Data Science**

Coursework: Computer Architecture, Algorithms, Web Development, Database Design, Machine Learning

Skills

Languages: Java, JavaScript, Python, SQL

Frameworks & Libraries: Vue.js, Node.js, Pandas, NumPy, Leaflet

Tools & Technologies: Git, GitHub, Docker, Linux, REST APIs, Firebase, Agile/Scrum, Vercel, Render

Software Engineering Projects

Minnesota Alliance on Crime (Nonprofit) | MN Crime Resource Directory **February 2026 – August 2026**

Frontend Engineer | UI/UX | QA

- Owned frontend, UI/UX, and QA for a nonprofit crime resource directory containing 370+ Minnesota resources
- Expanded initial scope within Agile workflow by leveraging sprint and backlog management to pitch and build 5+ unprompted technical implementations, owning all frontend decisions
- Implemented test cases and performed cross-browser and cross-device QA for full-stack integration across Chrome, Firefox, Edge, mobile phones, and tablets
- Validated and cleaned 300+ rows of resources, fixing data inconsistencies, misspellings, and broken links
- Engineered map rendering based on 3 edge cases depending on physical address availability
- Optimized mobile loading, reducing time for map and resource list population from 15s to 1s
- Built reusable design system with standardized UI, components, and typography
- Designed and implemented accessibility and safety considerations including quick exit button positioning, dark mode coloring, 24/7 crisis lines, and resizable data columns

Tech Stack: Vue 3, Vite, Leaflet, Firebase, Shepherd.js, CARTO, Vercel

FoodScale | Food Allocation & Macronutrient Calculator **June 2026 – November 2026**

Full Stack Engineer

- Developed and deployed full stack food allocation calculator for meal preparation, generating food portions based on nutritional targets and available quantities of ingredients
- Integrated external API of 300,000+ food items with debounced search and filter fallback handling to populate calorie and protein per 100g per ingredient
- Implemented greedy algorithm to generate plans with the most protein and least protein ways to fill out user-input calorie goals
- Blended both generations using linear interpolation to calculate plan that matches protein goals, outputting the complete meal plan and the error between input and output
- Built automated testing against 20 cases, consistently receiving minimum variations between results and goals

Tech Stack: Vue 3, Vite, Vitest, USDA FoodData Central, Render

GreenStep Cities (Nonprofit) | Sustainability Tracker **September 2025 – December 2025**

Database developer

- Collaborated to design 15-table relational schema for an internal sustainability-tracking application
- Developed table and queries with defined relationships and constraints to support backend integration

Tech Stack: MySQL

Additional Experience

Home Depot, Richfield, MN **June 2023 – October 2024**

Service Desk

- Assisted 50+ customers a day with order management, returns, and issue resolutions